

FILTER DESIGN TEST CASE

High-Pass Filter - GPS L5 (1176 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a High-Pass Filter filter targeting GPS L5 (1176 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on AIN on Si substrate with a target center frequency of 1387.9 MHz and bandwidth of 249.2 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: High-Pass Filter
- Frequency Band: GPS L5 (1176 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: AIN on Si
- Package: CSP (Chip Scale Package)
- Design Tool: COMSOL Multiphysics 6.2
- Design Center: Osaka Design Center

This specification applies to all variants of the High-Pass Filter design targeting GPS L5 (1176 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1387.9 MHz
Bandwidth (3 dB):	249.2 MHz
Insertion Loss Target:	≤ 1.4 dB
Return Loss Target:	≥ 23.0 dB
Out-of-Band Rejection:	≥ 46.0 dB
Isolation (Duplexer):	≥ 43.0 dB
Q Factor:	10098
Temperature Coefficient:	-10.3 ppm/°C
Die Size:	2.2 x 2.5 mm
Wafer Size:	8-inch
Number of Resonators:	10
Electrode Material:	W

Passivation: SiO2

4. TEST PROCEDURE

1. Open COMSOL Multiphysics 6.2 and load the High-Pass Filter template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 1387.9 MHz, BW 249.2 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on Si).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 4164 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.4 dB
- Return loss in passband shall be >= 23.0 dB
- Out-of-band rejection at +/- 374 MHz offset >= 46.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 1.8 MHz
- Power handling shall be >= +28 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 94%

6. TEST RESULTS

Measurement Date: 2024-08-04
Devices Tested: 38
Insertion Loss (meas): 1.12 dB
Return Loss (meas): 24.7 dB
Rejection (meas): 49.1 dB
Isolation (meas): 47.6 dB
Q Factor (meas): 10098
Group Delay Variation: 2.8 ns
Power Handling: +36 dBm
Die Yield: 87.1%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The High-Pass Filter design demonstrated acceptable performance for GPS L5 (1176 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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